

Antian Wang, Ph.D.

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Education

- 2018 – 2024 • **Ph.D. Electrical Engineering (Intelligent System), The Holcombe Department of Electrical and Computer Engineering, College of Engineering, Computing and Applied Sciences, Clemson University, SC, U.S.A**
Advisor: Dr. Yingjie Lao (Now with Tufts University)
Dissertation title: Hardware-oriented Protection and Acceleration for Machine Learning Application.
- 2018 – 2021 • **Engineering and Science Education Certificate, Department of Engineering and Science Education, College of Engineering, Computing and Applied Sciences, Clemson University, SC, U.S.A**
Eleven Credit Hour courses in Seminar, Pedagogy, Education Research Methods, Professional development, and Practicum.
- 2013 – 2017 • **B.E. Communication Engineering, Department of Electronic Engineering, College of Information Engineering, Shanghai Maritime University, Shanghai, China.**
Municipal level Outstanding Graduates.
Outstanding Individual of Construction of Study Style.

Research Interest

Design Automation, Cryptographic System Software-Hardware co-design, Privacy-Preserving Machine Learning, Machine Learning Application Hardware Acceleration, VLSI Digital signal processing, Embedded System, Hardware Security, Cybersecurity, Machine Learning Model Security, Engineering Education.

Experience/Employment

- 2024 Aug – current • **Assistant Professor of Computer Engineering, Department of Electrical and Computer Engineering, College of Engineering, Technology, and Computer Science. Purdue University Fort Wayne, Fort Wayne, IN, U.S.A**
- 2024 Jan – 2024 July • **Visiting Graduate Student, Department of Electrical and Computer Engineering, Tufts University, Medford, MA, U.S.A.**

Research Grant

- Collaborative Research: SHF: Small: Efficient and Scalable Privacy-Preserving Neural Network Inference based on Ciphertext-Ciphertext Fully Homomorphic Encryption - National Science Foundation (NSF): CCF-2243052 & CCF-2243053; April 2023 - March 2026; Total Award amount: \$600,000 (Helped my advisor in writing the proposal).

Research Publications

Journal Articles

- 1 **A. Wang**, W. Tan, Z. Xu, T. Wei, C. Ding, K. K. Parhi, and Y. Lao, "Hedwig: Homomorphic encryption accelerator design using bfv-hps with high-speed fixed-point approximation," *IEEE Transactions on Circuits and Systems I: Regular Papers*, 2026.

- 2 W. Tan, S.-W. Chiu, **A. Wang**, Y. Lao, and K. K. Parhi, "PaReNTT: Low-latency parallel residue number system and NTT-based long polynomial modular multiplication for homomorphic encryption," *IEEE Transactions on Information Forensics & Security*, vol. 19, pp. 1646–1659, 2024.
- 3 W. Tan, **A. Wang**, X. Zhang, Y. Lao, and K. K. Parhi, "High-speed vlsi architectures for modular polynomial multiplication via fast filtering and applications to lattice-based cryptography," *IEEE Transactions on Computers*, vol. 72, no. 9, pp. 2454–2466, 2023.
- 4 W. Tan, B. M. Case, **A. Wang**, S. Gao, and Y. Lao, "High-speed modular multiplier for lattice-based cryptosystems," *IEEE Transactions on Circuits and Systems II: Express Briefs*, vol. 68, no. 8, pp. 2927–2931, 2021.
- 5 **A. Wang**, W. Tan, Y. Wen, and Y. Lao, "NoPUF: A novel puf design framework toward modeling attack resistant PUFs," *IEEE Transactions on Circuits and Systems I: Regular Papers*, vol. 68, no. 6, pp. 2508–2521, 2021.

Conference Proceedings

- 1 K. Zhang, **A. Wang**, K. K. Parhi, and Y. Lao, "DRS: Unified fully homomorphic encryption hardware accelerator with scheme-switching," in *IEEE Asia and South Pacific Design Automation Conference*, IEEE, 2027.
- 2 **A. Wang**, W. Tan, Z. Xu, T. Wei, C. Ding, K. K. Parhi, and Y. Lao, "HEDWIG: Homomorphic encryption accelerator design using bfv-hps with high-speed fixed-point approximation," in *Proceedings of the 2025 ACM/SIGDA International Symposium on Field Programmable Gate Arrays*, 2025, pp. 184–184.
- 3 **A. Wang**, K. Zhang, K. K. Parhi, and Y. Lao, "RECAPHE: Reconfigurable polynomial modular computation architectures for unified PQC and HE schemes," in *2025 59th Asilomar Conference on Signals, Systems, and Computers*, IEEE, 2025, pp. 1288–1292.
- 4 **A. Wang**, K. Zhang, K. K. Parhi, and Y. Lao, "HERMES: Homomorphic encryption over residual number system for multi-level evaluations," in *IEEE International Conference on Computer-Aided Design*, IEEE, 2024.
- 5 K. Zhang, **A. Wang**, K. K. Parhi, and Y. Lao, "Hardware acceleration for fully homomorphic encryption scheme switching from CKKS to FHEW," in *2024 58th Asilomar Conference on Signals, Systems, and Computers*, IEEE, 2024, pp. 1792–1796.
- 6 **A. Wang** and Y. Lao, "Homomorphic evaluation friendly vision transformer design," in *2023 57th Asilomar Conference on Signals, Systems, and Computers*, IEEE, 2023, pp. 7–10.
- 7 **A. Wang**, B. Zhao, W. Tan, and Y. Lao, "NNTesting: Neural network fault attacks detection using gradient-based test vector generation," in *Proceedings of the 60th Annual Design Automation Conference 2023*, 2023, pp. 1–6.
- 8 **A. Wang**, W. Tan, and Y. Lao, "Integral sampler and polynomial multiplication architecture for lattice-based cryptography," in *2022 IEEE International Symposium on Defect and Fault Tolerance in VLSI and Nanotechnology Systems (DFT)*, IEEE, 2022, pp. 1–6.
- 9 W. Tan, **A. Wang**, Y. Lao, X. Zhang, and K. K. Parhi, "Pipelined high-throughput NTT architecture for lattice-based cryptography," in *2021 Asian Hardware Oriented Security and Trust Symposium (AsianHOST)*, IEEE, 2021, pp. 1–4.
- 10 W. Tan, **A. Wang**, K. K. Parhi, and Y. Lao, "Area-efficient pipelined VLSI architecture for polar decoder," in *2020 IEEE Computer Society Annual Symposium on VLSI (ISVLSI)*, IEEE, 2020, pp. 397–402.

Patents

- 1 K. K. Parhi, W. Tan, S.-W. Chiu, **A. Wang**, and Y. Lao, *Parallel polynomial modular multiplication using NTT and inverse NTT*, U.S. Patent Application 18/500,670, May 2023.

² K. K. Parhi, X. Zhang, W. Tan, **A. Wang**, and Y. Lao, *Low latency polynomial modulo multiplication over ring*, U.S. Patent Application 17/582,560, Jul. 2023.

Preprint/In preparation

¹ K. Zhang, M. S. Salek, **A. Wang**, M. Rahman, M. Chowdhury, and Y. Lao, "Preparing for kyber in securing intelligent transportation systems communications: A case study on fault-enabled chosen-ciphertext attack," 2025.

Professional Activities

Conference Reviewer

AIES	• AAAI/ACM Conference on AI, Ethics, and Society	2025
AsianHOST	• IEEE Asian Hardware Oriented Security and Trust Symposium	2021-2023
Asilomar	• Asilomar Conference on Signals, Systems & Computers	2025-2026
GLVISI	• IEEE/ACM Great Lakes Symposium on VLSI	2024
HOST	• IEEE International Symposium on Hardware Oriented Security and Trust	2023-2024
ISCAS	• IEEE International Symposium on Circuits and Systems	2019-2026
ISVLSI	• IEEE Symposium on VLSI	2022
MWSCAS	• IEEE International Midwest Symposium on Circuits and Systems	2020,2022-2026
SiPS	• IEEE International Workshop on Signal Processing Systems	2019-2020, 2024

Journal Reviewer

AJSE	• Arabian Journal for Science and Engineering
TCAS-I	• IEEE Transactions on Circuits and Systems I: Regular Papers
TCAS-II	• IEEE Transactions on Circuits and Systems Part II: Express Briefs • Journal of Computational Electronics
HaSS	• Journal of Hardware and Systems Security • The Journal of Supercomputing
OJCAS	• IEEE Open Journal of Circuits and Systems
PEER	• Peer-to-Peer Networking and Applications
TVLSI	• IEEE Transactions on Very Large Scale Integration Systems

Conference Session Chair

Asilomar	• Asilomar Conference on Signals, Systems & Computers	2025
DAC	• Design Automation Conference	2026

Technical Program Committee

	• IEEE Cloud Summit	2025-2026
IEEE Healthcom	• IEEE International Conference on E-health Networking, Application and Services Workshop	2026

Professional Activities (continued)

University Activities at Purdue University Fort Wayne

- Departmental Computer Engineering Curriculum Committee (2024-)
- Departmental Scholarship Selection Committee (2025 Spring)
- ETCS Assembly (2024-2027)
- ETCS Faculty Affairs Committee (2024-2027)
- ETCS Curriculum Committee (2025 Spring)
- ETCS Faculty Awards Selection Committee (2026 Spring)

Membership

- IEEE Member
- ACM Member
- ASEE Member

Teaching

Senior Engineering Design Advisor at Purdue University Fort Wayne

ECE 40500/40601	• Digital Signal Processing for Future Weather Satellite	2026 Fall-2027 Spring.
	Co-advised with Dr. David Cochran. Sponsored by Logistikos Engineering LLC.	
	• Real-Time Embedded Motor Control System for BLDC Fuel Transfer Pump	2025 Fall-2026 Spring.
	Co-advised with Dr. David Cochran. Sponsored by Fill-Rite Company.	
	• CO ₂ Incubator for Cell Culture	2024 Fall-2025 Spring.
	Co-advised with Dr. David Cochran. Sponsored by Bionanomics / MIE.	

Instructor at Purdue University Fort Wayne

ECE 27000	• Introduction to Digital System Design	2024 Fall/ 2027 Spring.
	• Introduction to Digital System Design with Lab	2026 Spring.
ECE 36200	• Microprocessor Systems and Interfacing	2025 Spring/2025, 2026 Fall.
ECE 43700	• Computer Design and Prototyping with Lab	2025, 2026, 2027 Spring.
ECE 48500	• Embedded Real-Time Operating Systems with Lab	2024, 2025, 2026 Fall.
ECE 56700	• FPGA Design For Signal Processing Applications	2027 Spring.

Teaching Grants at Purdue University Fort Wayne

AY25-26	• Students as Partners Grant for ECE 27000 Introduction to Digital System Design.
2026	• Faculty Academy on Excellence in Teaching (FACET) mini-grant for Scholarship of Teaching and Learning (SoTL) (\$120)
AY26-27	• Course Design for New(er) Faculty Grant for ECE 43700 Computer Design and Prototyping.

Teaching (continued)

Guest Lecturer

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| EE 0193 | • Special Topics: AI Security and Privacy, Tufts University | 2024 Spring. |
| | Privacy Preserving machine learning with an emphasis on the Homomorphic encryption approaches. | |

Graduate Teaching Assistant at Clemson University

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| ECE 2090 | • Logic and Computing Devices Laboratory | 2022 Fall-2023 Fall. |
| | Perform ABET evaluation material collection and provide suggestions for in-class Lab Kit preparation. | |
| | Change the simulation software to the latest Logisim Evolution and prepare the usage tutorial. | |
| | Perform and update all required lab circuit board building diagrams according to the syllabus. Listed observed common mistakes for future references. | |
| | Enrich lab capstone/final project topics with hints. | |
| | Enrollment: 71 students for 23 Fall, 35 students for 23 Spring, 41 students for 22 Fall. | |
| ECE 4590/6590 | • Integrated Circuit Design | 2023 Fall. |
| | Reformat HSPICE lab to using the Cadence SPECTRE tool. | |
| | Introduce Cadence RTL to GDSII workflow. | |
| | Enrollment: 25 students. | |

Graduate Grader Assistant at Clemson University

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| ECE 2010H | • Logic and Computing Devices | 2020 Fall. |
| ECE 2070 | • Basic Electrical Engineering | 2019 Fall – 2020 Fall. |
| ECE 2720 | • Computer Organization | 2020 Fall – 2021 Fall. |
| ECE 3300 | • Signals, Systems, and Transforms | 2022 Spring. |
| ECE 3710 | • Micro-controller Interfacing | 2020 Fall – 2021 Fall. |
| ECE 4090 | • Introduction to Linear System | 2019 Spring, 2019 Fall. |
| ECE 4270 | • Communication Systems | 2021 Spring - 2023 Spring. |
| | Prepare solutions and practice problems for Fall semesters. | |

Miscellaneous Experience

Conference Travel Grant

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| ISCAS 2022 | • ISCAS 2022 Student Participation Grants |
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Graduate Travel Grant (GTG) from Clemson University Graduate Student Government

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| 2023 Summer | • Design Automation Conference (DAC) 2023, San Francisco, CA. |
| 2022 Summer | • International Symposium on Circuits and Systems (ISCAS) 2022, Austin, TX, USA. |
| 2021 Fall | • Design Automation Conference (DAC) 2021, San Francisco, CA/Virtual, USA. |
| 2021 Spring | • International Solid-State Circuits Conference (ISSCC) 2021, Virtual, USA. |

Advising

Undergraduate Students

Honor Project	• Sankalp Sinha	2026 Spring
	“Cross-Language Analysis of Prompt Injection in Large Language Models Using Behavioral Detection”	
	Last known employment/ graduate school: Northwestern University	

Skills

Programming	• C/C++/C#, OpenCL, CUDA, Python, R, Matlab, Verilog HDL, VHDL, $\LaTeX$, SPICE, SPECTRE.
Tools	• TensorFlow, PyTorch, Xilinx Vitis, Xilinx Vivado, Cadence (XCELIUM, Genus, Innovus), Synopsys (VCS/Verdi, Design Compiler)